

# Software Requirements Specification (SRS)

## Smart Queue Management System

### 1. Introduction

The **Smart Queue Management System (SQMS)** is a web-based application designed to simplify and automate queue management in banks and other service organizations. The system enables customers to book queue tokens online, monitor their queue status in real time, and receive notifications. Staff members can efficiently manage customer queues, while administrators oversee branches, services, counters, and system operations. The system improves service efficiency, reduces waiting time, and enhances customer satisfaction.

### 2. Purpose

The purpose of the Smart Queue Management System is to provide an efficient, secure, and user-friendly platform for managing customer queues. It minimizes manual queue handling by allowing online token booking, QR code verification, real-time queue tracking, and automated notifications. The system helps customers save time, assists staff in managing queues effectively, and enables administrators to monitor and control overall operations.

### 3. Scope

The Smart Queue Management System covers the complete queue management process, from customer registration to service completion. Customers can register, log in, book tokens, track their queue status, receive notifications, and provide feedback. Staff members can verify customer tokens, call the next customer, update queue status, and complete services. Administrators can manage branches, services, counters, staff, reports, and system settings. The system integrates with Email APIs and QR Code APIs to provide secure and efficient service management.

### 4. Problem Statement

Traditional queue management systems rely heavily on manual token distribution and physical waiting lines, resulting in long waiting times, overcrowding, poor customer experience, and inefficient service management. Customers often have no information about their queue status or estimated waiting time. Staff members face difficulties in managing queues efficiently, while administrators lack real-time monitoring and reporting capabilities. These challenges highlight the need for an automated and intelligent queue management solution.

### 5. Proposed Solution

The proposed Smart Queue Management System provides a digital solution to automate the entire queue management process. Customers can book queue tokens online, receive QR codes for verification, track their queue position in real time, and receive email notifications regarding their bookings and service updates. Staff members can manage queues through a dedicated dashboard, while administrators can monitor system activities, generate reports, and manage organizational resources. This solution improves operational efficiency, reduces waiting time, and enhances customer satisfaction.

### 6. User Roles

#### Customer

- Register and log in
- Book queue tokens
- Track live queue status

- View booking history
- Receive notifications
- Submit feedback

#### **Staff**

- Log in securely
- View assigned counter
- Verify customer QR codes
- Call the next token
- Update token status
- Complete customer service

#### **Admin**

- Manage branches
- Manage services
- Manage counters
- Manage staff
- Monitor live queues
- Generate reports
- Configure system settings

## **7. Functional Requirements**

The system shall provide role-based authentication for customers, staff, and administrators. Customers shall be able to register, log in, book queue tokens, track queue status, and receive notifications. Staff shall verify QR codes, manage customer queues, and update token statuses. Administrators shall manage branches, services, counters, staff, and generate reports. The system shall also maintain booking history and customer feedback.

## **8. Non-Functional Requirements**

The system shall provide high performance with fast response times and support multiple concurrent users. It shall ensure security through encrypted passwords, secure authentication, and role-based access control. The application should be reliable, available 24/7, scalable for multiple branches, and easy to maintain. The user interface should be simple, responsive, and easy to navigate.

## **9. System Requirements**

#### **Software Requirements**

- Java (JDK 17 or above)
- Spring Boot
- PostgreSQL
- Maven
- IntelliJ IDEA/Eclipse
- Postman
- Git

#### **Hardware Requirements**

- Intel Core i3 Processor or above
- 8 GB RAM
- 256 GB Storage
- Internet Connection
- Windows/Linux Operating System

## **10. System Features**

- Role-based user authentication
- Online queue token booking
- QR code generation and verification
- Real-time queue tracking
- Estimated waiting time display
- Email notifications
- Customer feedback management
- Queue status updates
- Branch and staff management
- Report generation and analytics

## **11. Future Enhancements**

Future versions of the Smart Queue Management System may include a mobile application for Android and iOS, AI-based waiting time prediction, SMS notifications, digital payment integration, multi-language support, facial recognition for customer verification, and advanced analytics dashboards for improved decision-making.